

NMAM INSTITUTE OF TECHNOLOGY, NITTE
 (An Autonomous Institution affiliated to VTU, Belagavi)
Eighth Semester B.E. (Credit System) Degree Examinations
 May - June 2023

19EE8X10 - NON-CONVENTIONAL ENERGY SYSTEMS

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing Two full questions from Unit – I & Unit – II each and One full question from Unit – III.

Unit – I		Marks	BT*	CO*	PO*
1. a)	Define the following (i) declination angle (ii) Latitude of location (iii) solar azimuth angle.	6	L*1	1	1
b)	With a neat sketch explain the working principle of solar still	6	L2	2	1,2,6,7,8
c)	Determine the average value of solar radiation on a horizontal surface for June 22, at the latitude 10° N, if constants a and b are given as equal to 0.3 and 0.51 respectively $\bar{n}/N = 0.55$. Value of $I_{sc} = 1353 \text{ W/m}^2$ or 4871 kJ/m^2 and $MKS = 1165 \text{ kcal/hrm}^2$	8	L3	1	1
2. a)	Compare between the conventional and non-conventional energy sources.	6	L2	1	1
b)	With a neat sketch explain the working principle of solar Green House.	8	L2	2	1,2,6,7,8
c)	List the advantages and disadvantages of solar photovoltaic systems.	6	L1	2	1,2,6,7,8
3. a)	With a neat electrical circuit, explain Angstrom Pyrheliometer used for measuring the beam radiation.	8	L2	1	1
b)	Explain the advantages of flat plate collectors.	6	L1	1	1
c)	With a neat sketch explain the working of solar drier.	6	L2	2	1,2,6,7,8
Unit – II					
4. a)	With a neat diagram explain the necessity of energy storage unit.	6	L2	3	1,6,7
b)	Wind at 1 standard atmospheric pressure and 15°C has velocity of 15m/s. Calculate: (i) the total power density in the wind stream (ii) the maximum obtainable power density (iii) a reasonably obtainable power density	8	L3	3	1,6,7
c)	Explain the process of biomass to ethanol conversion.	6	L2	4	1,6,7
5. a)	Explain the necessity of storage of solar energy.	6	L2	3	1,6,7
b)	Explain the factors to be considered for the selection of site in WECS.	8	L2	3	1,6,7
c)	What are the factors affecting biogas generation?	6	L2	4	1,6,7
6. a)	What are the advantages of wind energy conservation systems?	4	L1	3	1,6,7
b)	Derive the expression for maximum power in wind.	10	L3	3	1,6,7
c)	With a neat sketch explain the working principle of JANATA model.	6	L2	4	1,6,7
Unit – III					
7. a)	With a neat sketch explain the construction and working principle of tidal power plant	8	L2	5	1,6,7
b)	Explain the necessary properties of an ideal ion exchange membrane electrolyte	6	L2	5	1,6,7
c)	Explain the electrolytic production of hydrogen	6	L2	5	1,6,7
8. a)	With a neat diagram, explain the working principle of single basin tidal power plant.	8	L2	5	1,6,7
b)	Explain the merits and demerits of fuel cell	6	L2	5	1,6,7
c)	Explain the hydrogen production by the iron process	6	L2	5	1,6,7

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
